

EEE 308L

Electrical Circuits II Laboratory

Lab Report



Section: 06

Group No: 04

Experiment no: 02

Name of the experiment: Study of Operational Amplifier as Comparator.

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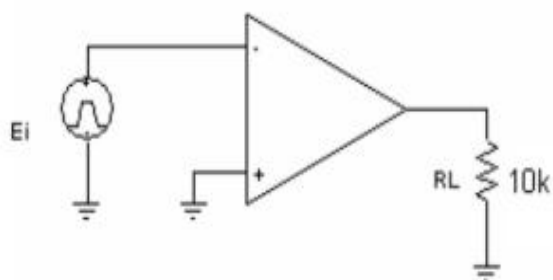
Experiment 01: Study of Operational Amplifier as Comparator.

Objective: To study op-amp as a comparator.

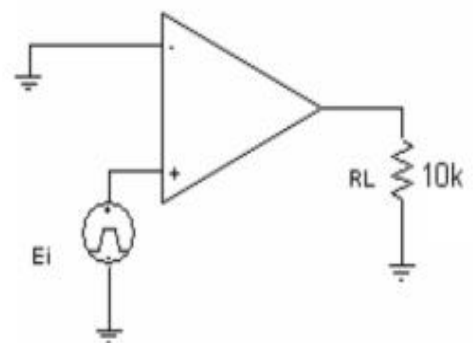
Equipment:

- IC: 741 1 Piece
- Resistor: 10k 1 Piece
- AT-7000 Trainer Board 1 Unit
- DC Power Supply 1 Unit
- +15 volts, -15 Volts
- Function Generator 1 Unit
- Oscilloscope 1 Unit
- Multimeter 1 Unit
- Wires

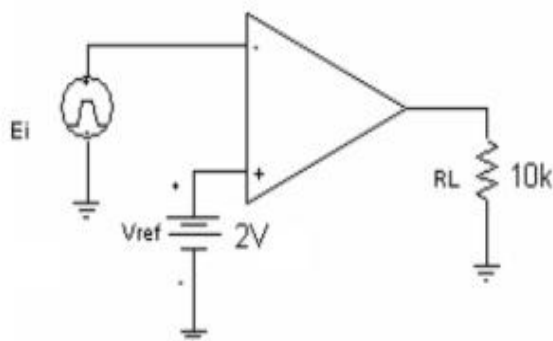
Circuit Diagram:



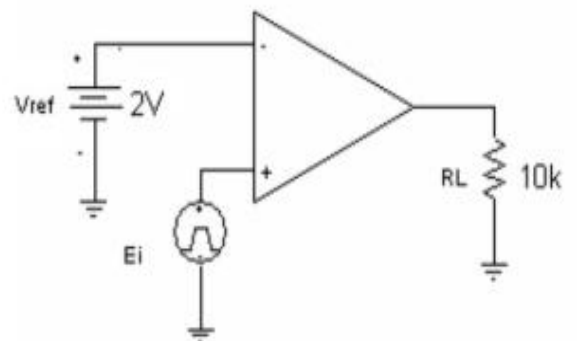
Inverting Zero-Crossing Detector



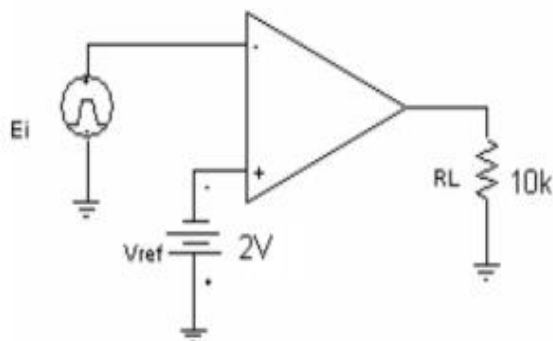
Noninverting Zero-Crossing Detector



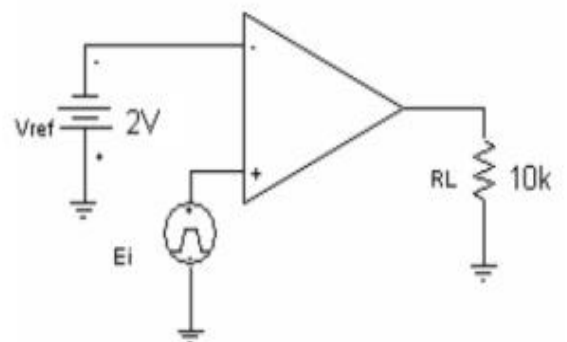
Inverting Positive-Voltage-Level Detector



Noninverting Positive-Voltage-Level Detector



Inverting Negative-Voltage-Level Detector



Noninverting Negative-Voltage-Level Detector

Experimental Set up:

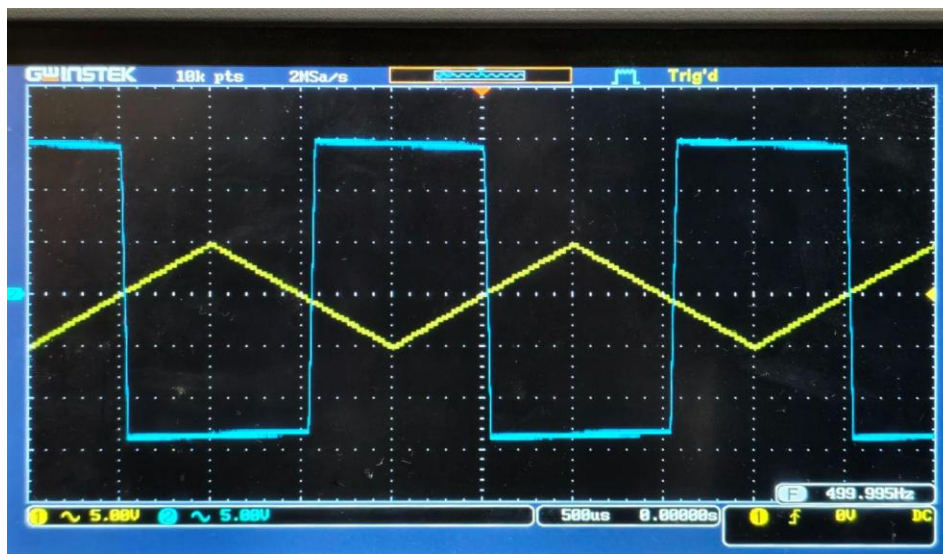
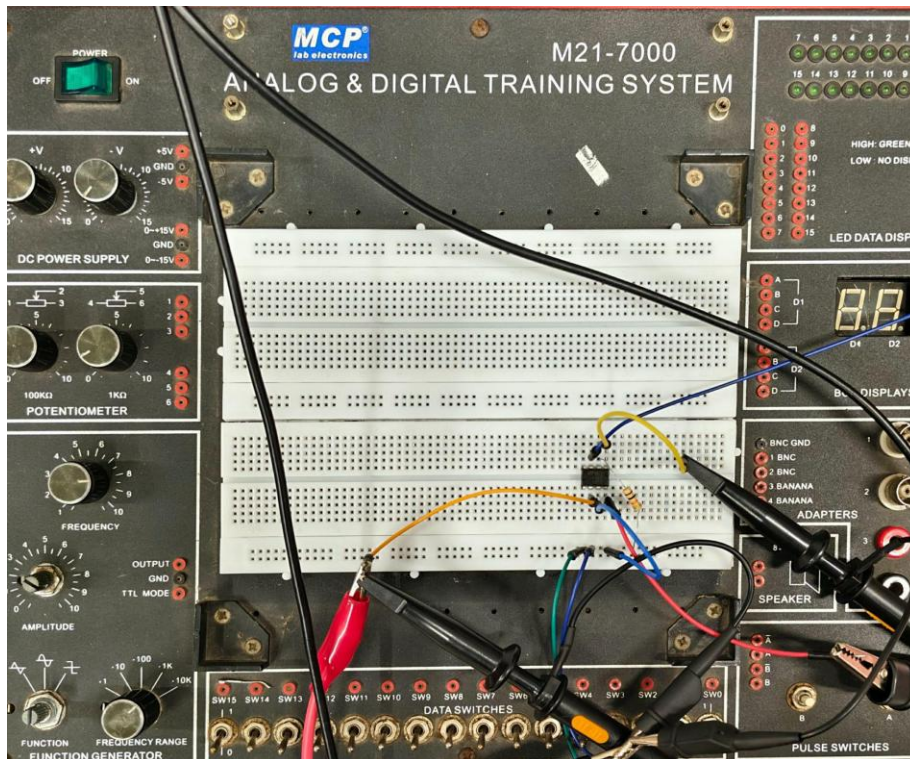


Fig-1: Inverting Zero-Crossing Detector Circuit And Input-Output Signal.

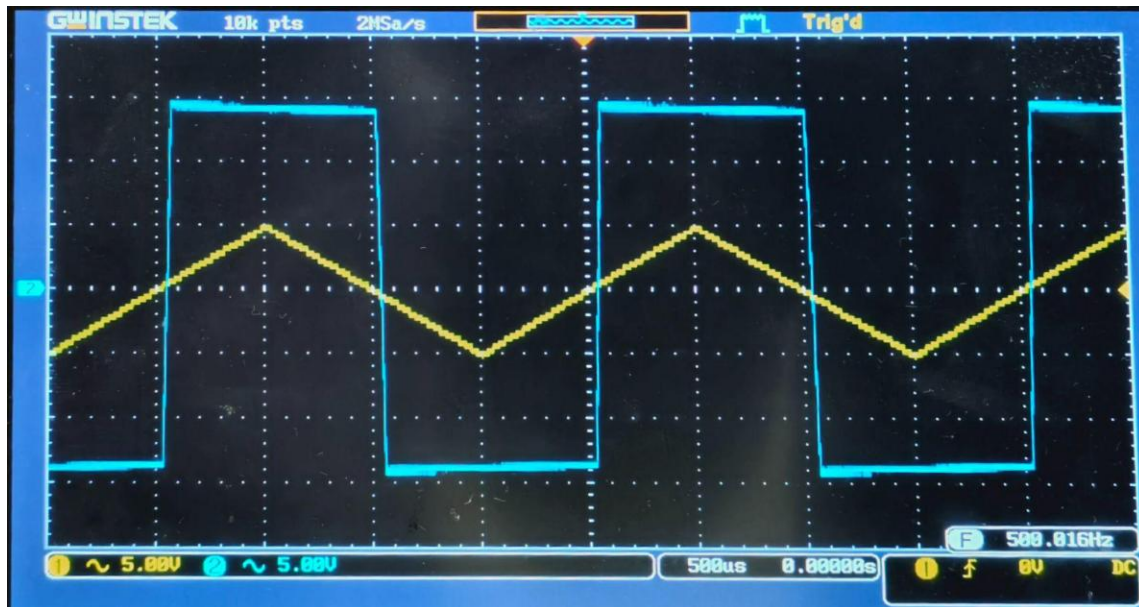
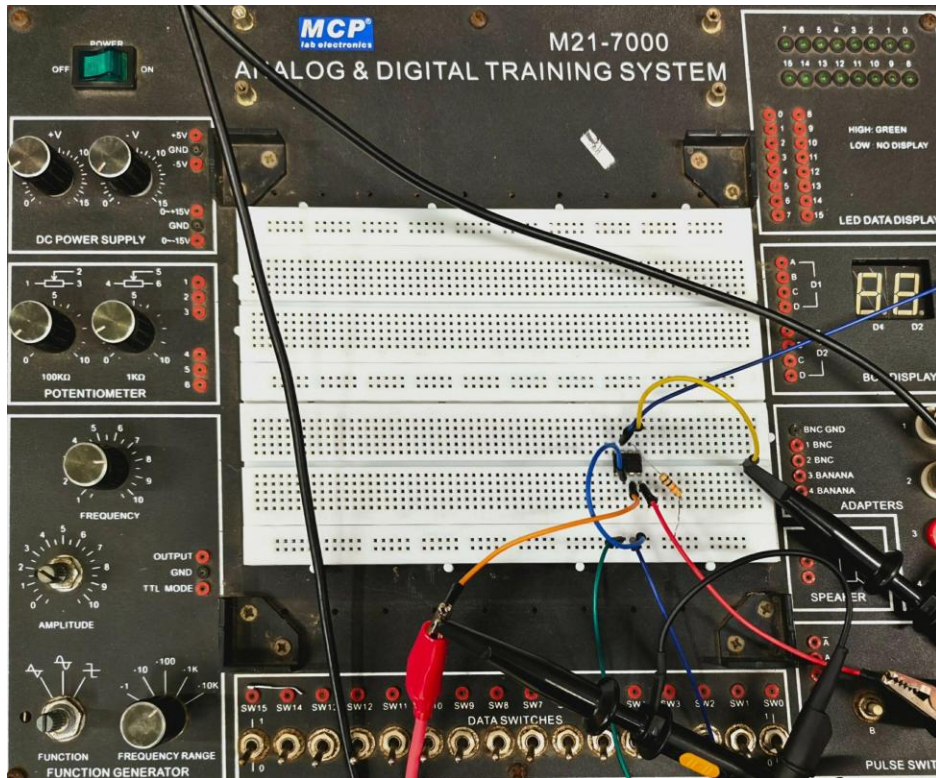


Fig-2: Non-Inverting Zero-Crossing Detector Circuit And Input-Output Signal.

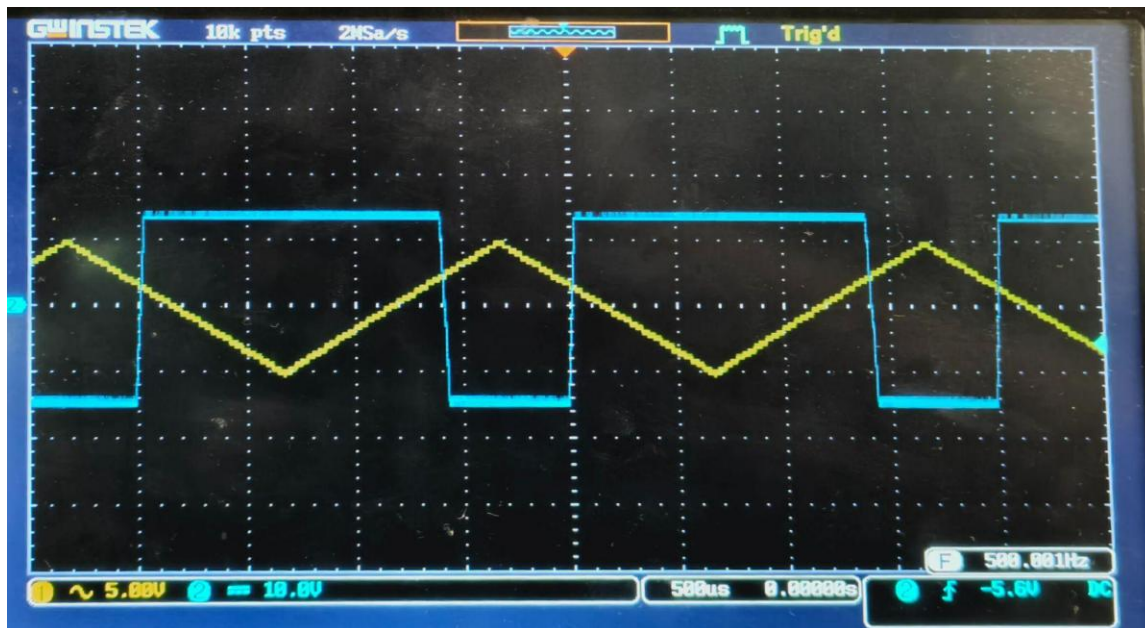
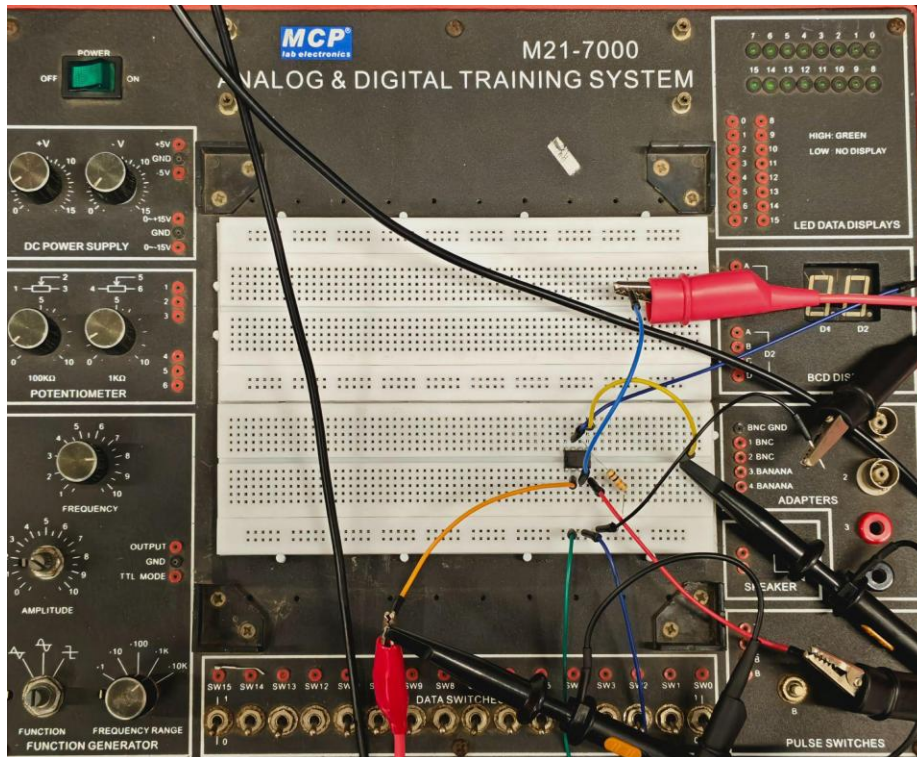


Fig-3: Inverting Positive-Voltage-Level Detector Circuit And Input-Output Signal

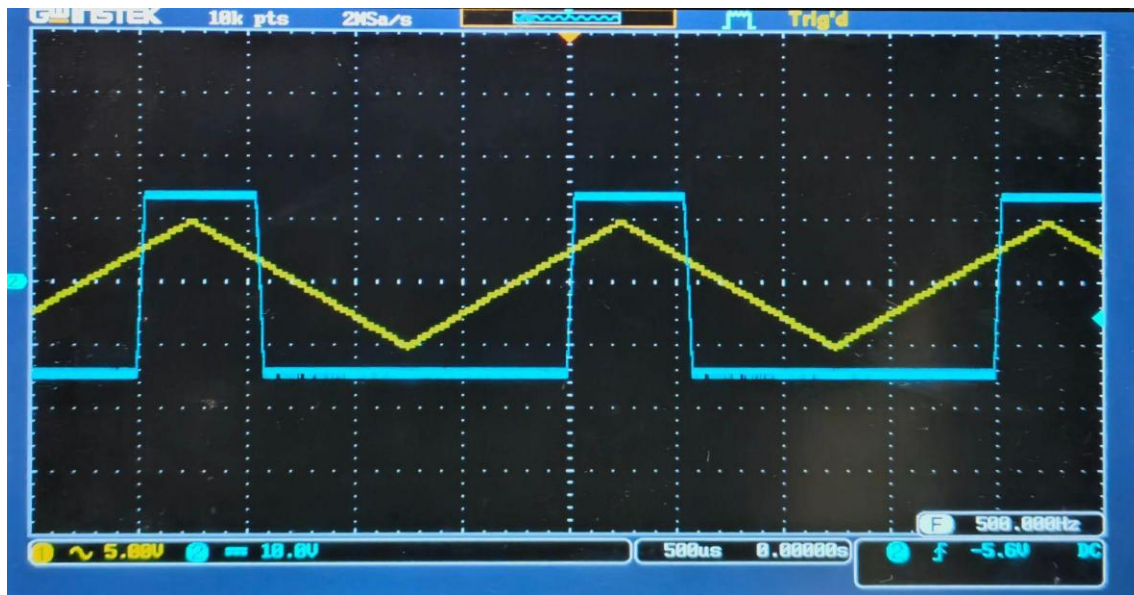
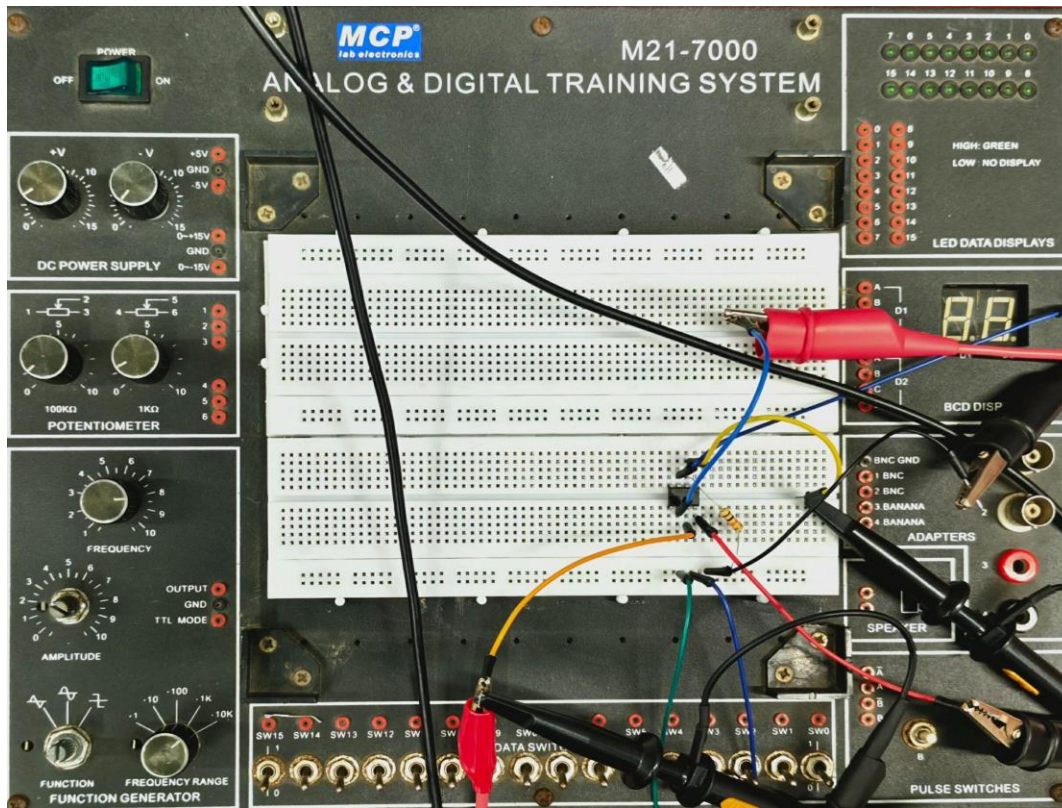


Fig-4: Non-Inverting Positive-Voltage-Level Detector Circuit And Input-Output Signal

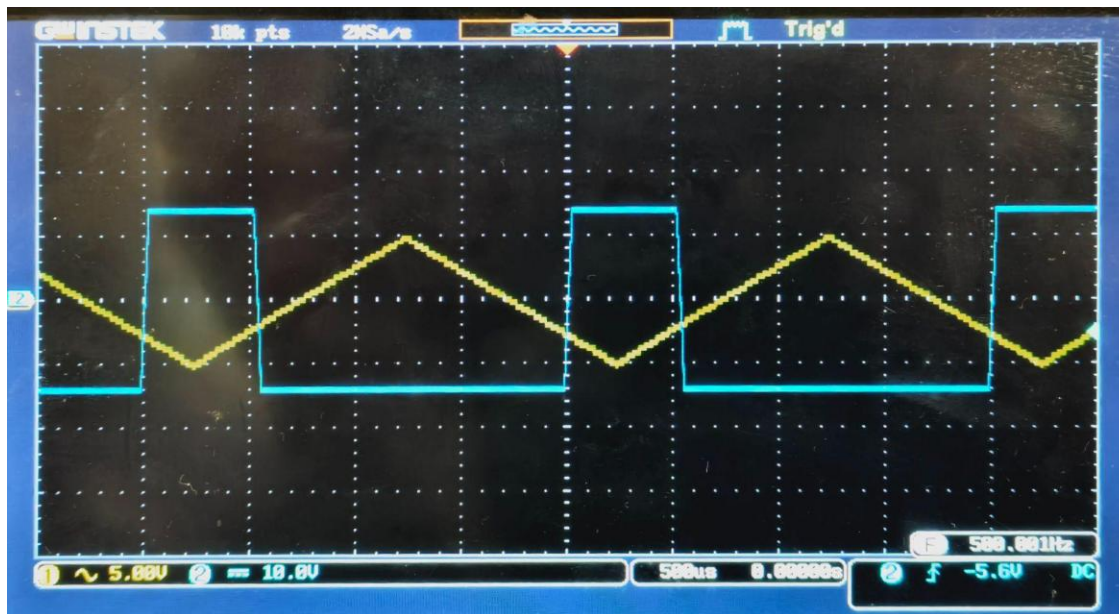
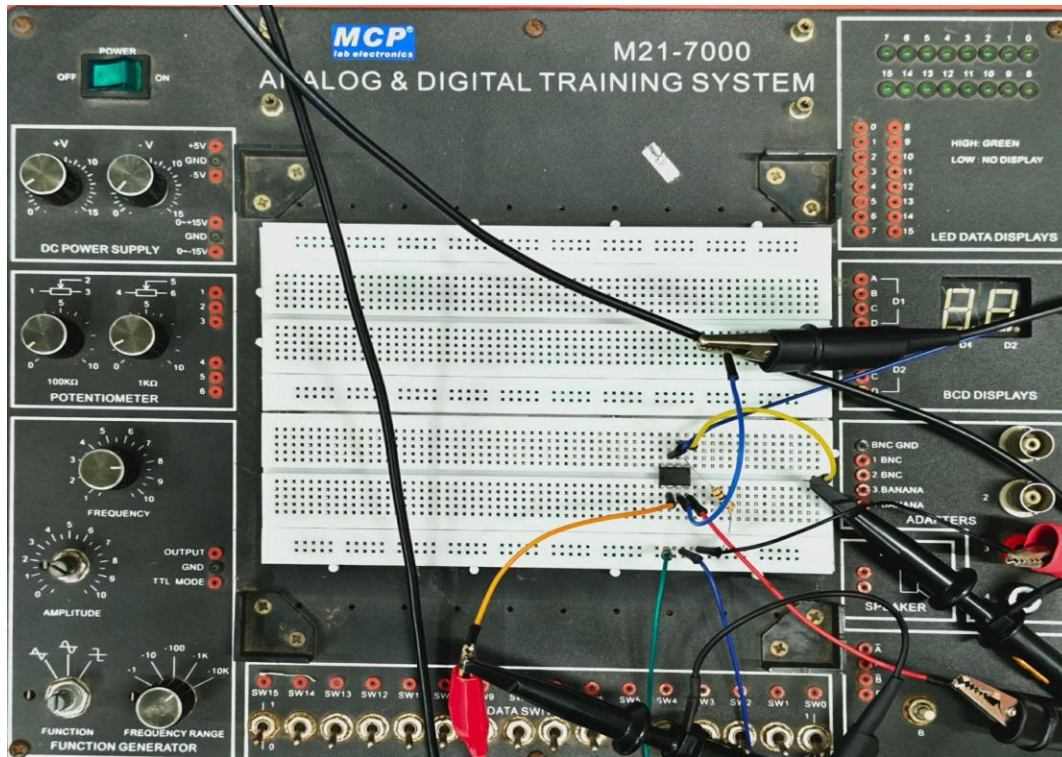


Fig-5: Inverting Negative-Voltage-Level Detector Circuit And Input-Output Signal

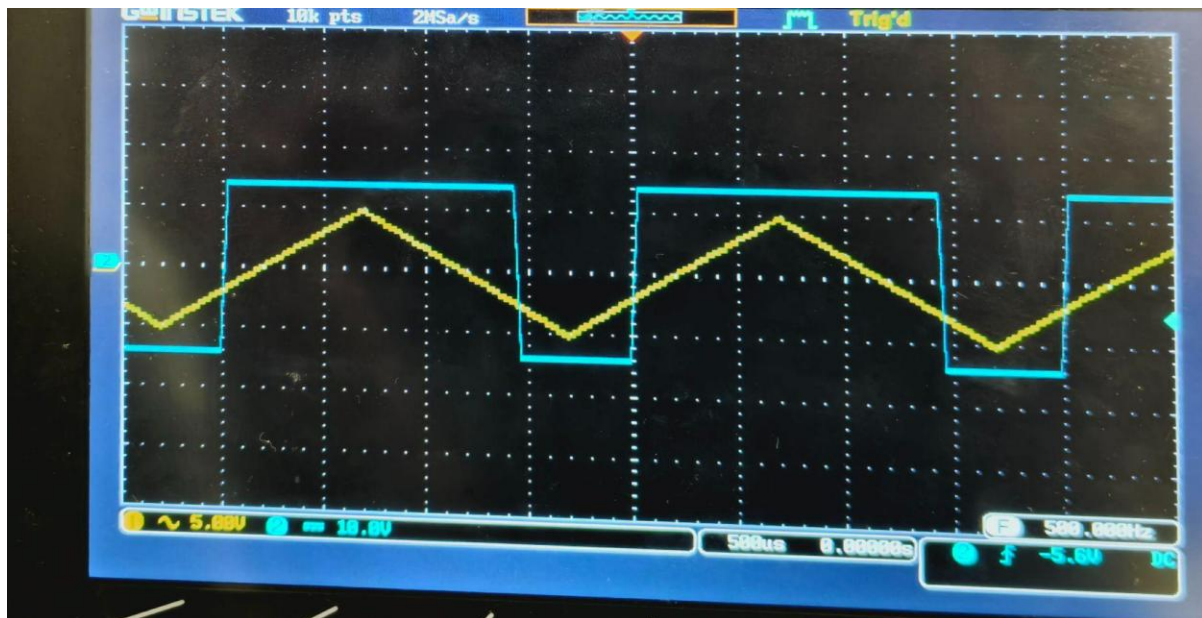
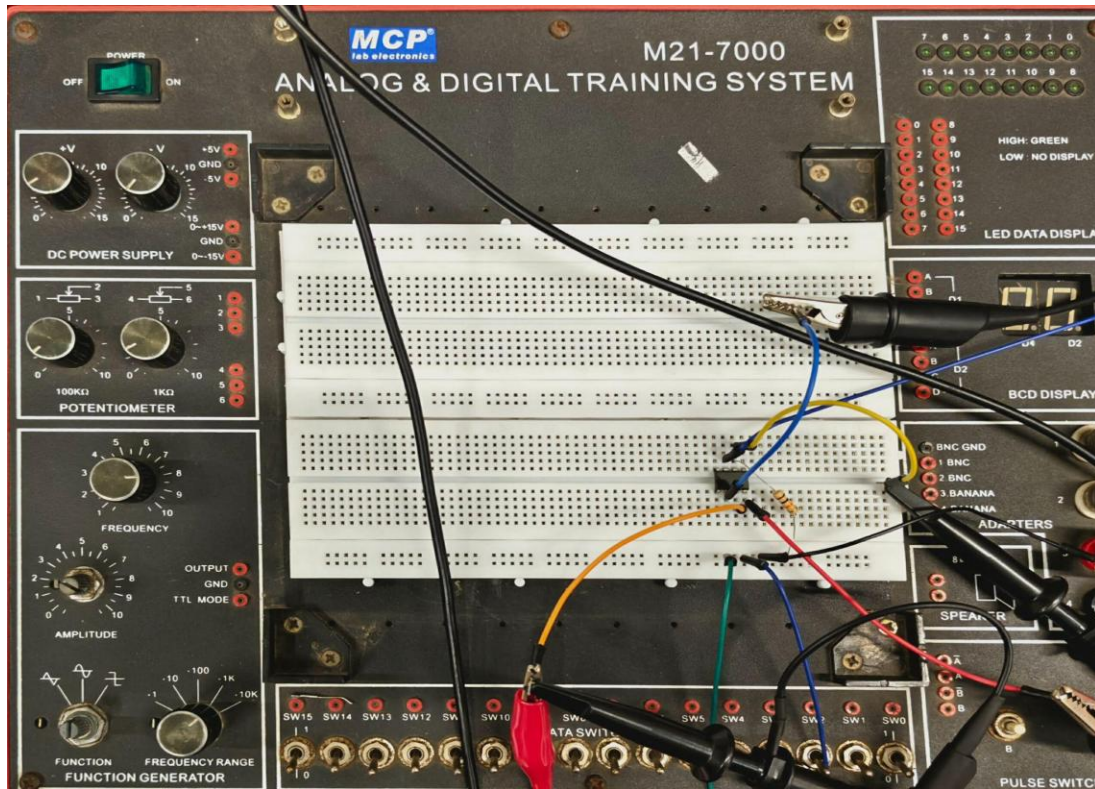


Fig-6: Non-Inverting Negative-Voltage-Level Detector Circuit And Input-Output Signal

Conclusion:

This experiment clearly demonstrated how an operational amplifier can function as a voltage comparator. When used in an open-loop configuration, the op-amp's output quickly switched between its positive and negative saturation levels. A triangular wave input was compared with a fixed reference voltage. In the case of zero-crossing detectors, the output shifted state exactly when the input passed through 0 V, effectively converting the triangular waveform into a square wave. For level-detection circuits, the output changed precisely when the input voltage exceeded the set positive or negative reference levels.

The observed behavior aligns perfectly with comparator theory. The op-amp's immense open-loop gain amplifies even the slightest differential input, forcing the output to its supply rail limits. This confirms the op-amp's effectiveness as a high-speed, decision-making circuit.